



## Handout 6.6: Method Match and Demonstration Record

Name: \_\_\_\_\_ Date: \_\_\_\_\_ Period: \_\_\_\_\_

Print pages 1 to 3 for every student. Page 4 is the method card set, one per group of four, printed and cut apart.

## Part 1. Two families of heat

Fill in the nine methods and the tool or equipment each one needs.

**DRY HEAT: no water touching the food. The surface dries out, so it can brown.**

| Method | What it is | Tool or equipment |
|--------|------------|-------------------|
| Bake   |            |                   |
| Roast  |            |                   |
| Broil  |            |                   |
| Toast  |            |                   |
| Sauté  |            |                   |

**MOIST HEAT: water or steam surrounds the food. The surface stays wet, so it cannot brown.**

| Method | What it is | Tool or equipment |
|--------|------------|-------------------|
| Boil   |            |                   |
| Simmer |            |                   |
| Steam  |            |                   |
| Poach  |            |                   |

**Star the three methods you have used yourself.**

**Note on frying:** we do not fry in this room. Frying needs oil at 350 degrees or more, and a pot of hot oil is the one hazard this course does not take on. Everything a fryer does, an oven or a broiler can get close to.

## Part 2. Browning, in your own words

When the surface of food gets hot and dry, \_\_\_\_\_



Water on the surface holds the temperature down, so nothing browns until \_\_\_\_\_.

One browned food I like: \_\_\_\_\_

What it tastes like compared to the unbrowned version:



### Part 3. Match eight foods to a method

For each food, pick one method and give a reason that says **what the heat does to that food**. Two of these have more than one good answer.

| Food                       | Best method | Reason: what the heat does |
|----------------------------|-------------|----------------------------|
| 1. A chicken breast        |             |                            |
| 2. A potato you want crisp |             |                            |
| 3. Broccoli                |             |                            |
| 4. Rice                    |             |                            |
| 5. An egg                  |             |                            |
| 6. A slice of bread        |             |                            |
| 7. Apple slices            |             |                            |
| 8. A pound of dry pasta    |             |                            |

**The two we could not agree on:** \_\_\_\_ and \_\_\_\_

Why: \_\_\_\_\_

## Part 4. The demonstration record

The food: \_\_\_\_\_ Method A: \_\_\_\_\_ Method B: \_\_\_\_\_

| What I observed                        | Method A | Method B |
|----------------------------------------|----------|----------|
| Color                                  |          |          |
| Texture (how it feels, inside and out) |          |          |
| Smell                                  |          |          |
| Taste (only if you chose to taste)     |          |          |

### The compare sentence

You must use the words **browning** and **moisture**.

The \_\_\_\_\_ one is \_\_\_\_\_ because \_\_\_\_\_





## Exit card

Dry heat is for \_\_\_\_\_, moist heat is for \_\_\_\_\_, and the food I would cook tomorrow is \_\_\_\_\_ by \_\_\_\_\_.

## Page 4: Method cards

Print one set per group of four and cut apart. Each card has the method name and a small picture.

|               |              |              |
|---------------|--------------|--------------|
| <b>BAKE</b>   | <b>ROAST</b> | <b>BROIL</b> |
| <b>TOAST</b>  | <b>SAUTÉ</b> | <b>BOIL</b>  |
| <b>SIMMER</b> | <b>STEAM</b> | <b>POACH</b> |

Grade 8 stretch, two extra cards:

|              |            |
|--------------|------------|
| <b>GRILL</b> | <b>FRY</b> |
|--------------|------------|

## Grade 8 stretch

1. Where do **grill** and **fry** belong, dry or moist? Why?

Grill: \_\_\_\_\_ Because \_\_\_\_\_

Fry: \_\_\_\_\_ Because \_\_\_\_\_

2. We skip both in this room. Give the real reason for each, not “it is dangerous.”

Grill: \_\_\_\_\_

Fry: \_\_\_\_\_

3. Name one food that uses **both** families in one recipe. Say the order and the reason for the order.

Food: \_\_\_\_\_

Order: first \_\_\_\_\_, then \_\_\_\_\_

Reason: \_\_\_\_\_

## Teacher key

### Part 1:

#### Dry heat

| Method | What it is                                                                                            | Tool                                    |
|--------|-------------------------------------------------------------------------------------------------------|-----------------------------------------|
| Bake   | Surrounded by hot dry air in an oven, usually for batters, doughs, and casseroles                     | Oven, sheet pan or baking dish          |
| Roast  | The same hot dry air, usually at a higher temperature, for meat and vegetables, so the outside browns | Oven, sheet pan or roasting pan         |
| Broil  | Heat from above only, very close and very hot, for a minute or two                                    | Oven broiler, broiler pan               |
| Toast  | Dry heat on the surface of bread or nuts or seeds until brown                                         | Toaster, toaster oven, or a dry skillet |
| Sauté  | A small amount of fat in a hot pan, food moved constantly, done fast                                  | Skillet, spatula or tongs, burner       |

#### Moist heat

| Method | What it is                                                                                      | Tool                                |
|--------|-------------------------------------------------------------------------------------------------|-------------------------------------|
| Boil   | Fully covered in water at a rolling boil, 212 degrees, big bubbles breaking the surface         | Pot, burner, colander               |
| Simmer | Same water, lower heat, small bubbles at the edge, for long cooking without breaking food apart | Pot, burner, wooden spoon           |
| Steam  | Food sits above boiling water and is cooked by the steam, never touching the water              | Pot with a steamer basket and a lid |
| Poach  | Cooked gently in liquid below a simmer, barely moving, for delicate food like eggs and fish     | Shallow pan, slotted spoon          |

**Sauté is a dry method.** Fat is not water, and fat gets hotter than water can, which is exactly why food browns in it. This is the most common student error in the sort and it is worth thirty seconds.

**Part 2, browning:** when the surface gets hot and dry, the sugars and proteins in the food change into new compounds that are brown and that smell and taste stronger. Nothing browns until the surface water is gone, because water holds the surface at about 212 degrees and browning needs hotter than that.

**Part 3, the match:**

| Food                     | Method                                       | Reason                                                                                                                                                                                 |
|--------------------------|----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Chicken breast        | Bake or roast (accept sauté)                 | Dry heat browns the outside while the inside cooks through; it needs to reach a safe internal temperature                                                                              |
| 2. Potato you want crisp | Bake, roast, or broil                        | Dry heat drives off surface water so the outside can brown and crisp                                                                                                                   |
| 3. Broccoli              | Steam (accept sauté or roast)                | Steam cooks it fast without leaving it sitting in water, so it stays firm and green and keeps more vitamins. Roasting browns it, which some students prefer, and that is a good answer |
| 4. Rice                  | Simmer or boil                               | Rice has to absorb water to soften. Only water surrounding it does that                                                                                                                |
| 5. Egg                   | Boil, poach, or sauté (scrambled)            | This is the intentionally arguable one. All three are correct with a reason                                                                                                            |
| 6. Slice of bread        | Toast (accept broil)                         | Dry heat, no water, so the surface browns and gets crunchy                                                                                                                             |
| 7. Apple slices          | Bake or sauté (accept simmer for applesauce) | The second intentionally arguable one. Baking or sautéing browns the sugar; simmering breaks it down into sauce                                                                        |
| 8. Pound of dry pasta    | Boil                                         | Pasta must absorb water to soften, and it needs a lot of moving water so it does not                                                                                                   |

|  |  |       |
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|  |  | stick |
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**The two designed to be argued about are 5 (egg) and 7 (apple).** Let the groups fight about them and do not resolve it; both have more than one right answer with a reason.

**Part 4, the compare sentence, sample:** the baked one is brown and crisp on the outside and dry and fluffy inside because the dry oven air pulled the moisture off the surface and let it get hot enough for browning, while the boiled one stayed pale and wet because it sat in moisture the whole time and the surface never got above the temperature of the water.

**Grade 8 stretch:**

1. Grill is dry heat (radiant heat and hot metal, no water, and it browns hard). Fry is dry heat, which surprises students. Oil is not water, and food fried in oil browns, which is the giveaway.
2. Grill: there is no grill in the room and an open flame indoors is not allowed. Fry: a pot of oil at 350 degrees or hotter is the single worst burn risk in a kitchen, oil fires cannot be put out with water, and a room full of middle schoolers moving around a pot of hot oil is a risk no lesson objective is worth.
3. Sample answers: a pot roast or a braise, seared first in a hot pan (dry) then cooked in liquid (moist), because the searing builds the brown flavor that liquid can never create and the liquid then makes a tough cut tender. Or mac and cheese: pasta boiled (moist) then baked (dry) so the top browns. Or a roasted then simmered tomato sauce.

**Scoring:** the handout is daily work, complete, partial, or missing. Look for nine methods in the right family with a tool each, all eight foods matched with a reason about the heat, and a compare sentence that uses both required words.